

# **Toward Data-Driven Anti-Fraud Governance: Anomaly Detection and Composite Risk Scoring for Port-Level Oversight in Digital Maritime Services**

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## **ABSTRACT**

Digital transformation in public services has reduced opportunities for conventional fraud. However, this measure does not fully eliminate the risk, instead it created new risks associated with process manipulation and irregular approval timing. This study addresses this challenge of detecting such risks within high transaction volume in digital public service. This research proposed a data driven anomaly detection framework through composite fraud risk screening index, functioning as an early-warning and prioritization mechanism rather than a fraud confirmation tool. By utilising vessel arrival approval (PKK) from 257 ports during 2025, the study integrated three complementary analytical approaches, including rule based, statistical analysis, and isolation forest as machine learning. From 734,622 approval data, the rule-based algorithm identified 56,306 transactions or 7.98% red flag condition based on predefined criteria. Moreover, modified z-score indicates 465 transactions as negative residual deviations beyond the threshold at -2.5, while isolation forest further identified multidimensional anomalous patterns based on approval duration, gross tonnage, and approval time. A composite Fraud Risk Screening Index (CFRSI) at the port level has been integrated based on three previous approaches with Lahewa, Samarinda, Pomako, Sinabang, and Benete are identified as the highest composite scores. The resulting index across all ports are then classified into five relative risk categories. As the identified anomalies represent operational red flags rather than confirmed fraud cases, further field-based validation is required before the findings can inform formal fraud determinations. The proposed framework contributes to anti-fraud governance by transforming high volume digital transaction into quantitative risk intelligence that can support risk-based prioritization of supervisory resources. In support of transparent and reproducible research, analytical workflow and source code are publicly available through github repository: <https://github.com/rifkiwijaya12/inaportnetRisk>.

**Keywords:** Fraud risk, anomaly detection, digital services, composite index, anti-fraud governance, maritime governance.

## **INTRODUCTION**

In the rapid advancement of digital technology, transformation has become an evitable process, unexceptionally government sector encouraged to adopt electronic system to improve administrative efficiency, transparency, and accountability. Within maritime sector, Indonesian Ministry of Transportation has introduced inaportnet platform to accelerate port clearance and vessel approval process. By replacing conventional manual procedures with digital workflows, the system is projected to reduce administrative duration from twenty four hours to approximately ten minutes (Admin, 2018). Furthermore, the platform can reduce direct interactions between service users and public officers, thereby mitigating opportunities for

bribery, unofficial payments, and other forms of administrative misconduct. However, the implementation of digital systems does not automatically eliminate fraud risks. It may transform to another fraudulent behaviour such as data manipulation, collusion among authorized actors, or exploitation of procedural loopholes. Consequently, digital workflows without implementing data-driven monitoring mechanisms risks masking fraudulent practices behind the speed and compliance of digital process.

Beyond the digital platform consequences, the operational scale of inaportnet introduces another critical issue for fraud oversight. The inaportnet system process more than one million vessel clearance transactions annually across all Indonesian ports, involving numerous shipping agents and port authorities. Traditional manual oversight of individual approval records becomes fundamentally impractical. Currently, supervisory practices remain largely focused on administrative compliance and service performance indicators rather than systematic fraud risk detection. Therefore, the absence of systematic and automated anomaly detection mechanisms encourage government to create fraud detection mechanisms that capable of continuously monitoring large-scale transactional datasets. The digital footprints generated by inaportnet represent an underutilized asset that can be converted into predictive anti-fraud intelligence. By deploying statistical and machine learning models, this method can transform administrative transactional data into quantifiable risk metrics, regulatory bodies can shift from reactive compliance to proactive risk mitigation.

Despite the growing interest on data analytics in public administration, existing literature on fraud detection in digital maritime sector remain underexplored. A systematic literature review by Alfian, Ritchi, and Adrianto (2023) of 43 studies on fraud analytics in public sector transaction have showed that implementation of anti-fraud analytic is overwhelmingly concentrated in developed nations and optimization of detection capabilities have left behind in prevention. Furthermore, existing studies on maritime sector primarily emphasize port efficiency, logistics performance, and digital transformation. Comparatively, fraud risk detection within digital workflows has a little attention. To address these gaps, this study develops a composite fraud risk screening index at port level for the inaportnet approval process by integrating multiple process derived from administrative transaction data. The proposed index function provides a data-driven early warning mechanism designed as a risk screening and prioritization tool, to identify operational patterns that deviate from expected norm (red flags) rather than as mechanism for confirming the occurrence of fraud. Therefore, this study offers a practical contribution by extending fraud risk analytics research into the maritime and port governance sector.

## **LITERATURE REVIEW**

According to Association of Certified Fraud Examiners (ACFE), as a leading international professional anti-fraud organization focused on preventing, identifying, and investigating fraudulent activities, fraud has been traditionally understood as an intentional deception committed to obtain unlawful or unethical benefits (ACFE, 2026). Similar document outlined about fraud tree which provides a comprehensive classification of all occupational fraud schemes, including corruption, asset misappropriation, and fraudulent financial statement. The impact of fraud is often associated with financial loses. However, the concept of fraud has evolved considerably within the context of public administration. This unethical behaviour often leads to not only financial damage of public institution, but also the erosion of societal trust (Supriadi, Tjakrawala, Suryadnyana, Sjam, & Marota, 2025). In extreme cases, fraudulent practices impact on poverty and cause human suffering (Kulmie, 2026).

The evolution of public service has transformed administrative workflows from physical or paper-based schemes to digital forms. This development is expected to reduce opportunities for traditional forms of corruption by minimizing direct interactions. However, this measure still fails to address completely persistent fraudulent behaviour. A study in China conducted to companies from 2007 to 2022, mentioned that digitalization has elevated both risk and frequency of fraudulent activities within corporations (Lu, Zhao, Xing, & Liu, 2025). Therefore, digital transformation alters the fraud mechanism from relying on document forgery, physical manipulation, fraud migrates toward exploitation of digital business process.

Unlike financial fraud detection that primarily targets unauthorized financial transaction (Prusti, Kumar, Purusottam, & Rath, 2021), in digital public services fraud detection increasingly focuses on identifying anomalies within administrative processes. Anomaly process may appear on unusually short or excessively long approval durations, irregular approval sequences, or deviations from established operational procedures. Given that fraudulent behaviour is progressively integrated into administrative processes, identification of fraud risks demands systematic analytical frameworks. The literature on fraud analytics and anomaly detection has developed several methodological approaches including rule/knowledge based (Ghorbani, Lu, & Tavallaei, 2010), statistical models (Alghushairy, Alsini, Soule, & Ma, 2021), and machine learning (Portela, Almenares Mendoza, & Benavides, 2019). Understanding the strengths and limitations of these approaches is essential for selecting an analytical framework that is appropriate for fraud risk detection within the operational context of the inaportnet system.

### 1. Rule based method

Rule based anomaly detection represents one of the most widely used for identifying irregularities within digital administrative approaches. This approach is constructed based on predefined rules obtained from patterns of normal and abnormal behaviour (Kraiem, Ghazzi, Peninou, Roman-Jimenez, & Teste, 2021). The rule-based method has several advantages including transparent, ease of implementation, and readily interpretable by auditors, supervisors and decision makers. Despite these advantages, anomaly detection utilised rule-based have several limitations that should be considered, such as accuracy of predefined business rule, no adapt with operational change, and unable to capture complex interactions of multiple indicators.

### 2. Statistical outlier detection

Given the limitation of previous anomaly detection methods, statistical detection can be considered to address former approach. Instead of relying on manually defined thresholds, statistical methods evaluate anomalies objectively from historical operational dataset. The modified z-score has been widely utilised to detect outliers as statistical analysis and applied in various fields, such as time series analysis. This characteristic is particularly relevant for administrative transaction data generated by digital public service systems. In addition, the performance of this method is determined by the specific scale of estimator such as Median Absolute Deviation (Yaro, Maly, Prazak, & Maly, 2024). Consequently, some estimators may perform better in certain context, while worse in others.

### 3. Unsupervised machine learning

Apart from statistical techniques, data analytic have expanded to identify complex patterns within large scale dataset. Unsupervised machine learning has been widely used in diverse fields such as fraud detection, healthcare, and industrial monitoring. This method has been

adopted for anomaly detection due to its capability without required labelled data (Catelani et al., 2024). Consequently, unsupervised methods often have lower accuracy and suffer from high false positive rates (Darban, Webb, Pan, Aggarwal, & Salehi, 2025). Furthermore, interpretation of the result has certain challenges due to the understanding of why certain data are flagged as anomalies.

Table 1 - Comparative of Anomaly Detection Approaches

| Approach                      | Main Principle                                                                                                    | Strengths                                                                                                                                                                  | Limitations                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rule-Based Method             | Detects anomalies based on predefined business rules, regulations, and operational thresholds.                    | a. high interpretability;<br>b. transparent on decision making;<br>c. easy implementation;<br>d. consistent with regulatory compliance;<br>e. suitable for audit purposes. | a. depends on predefined rules;<br>b. difficult to detect emerging fraud patterns;<br>c. limited adaptability to operational changes.                       |
| Statistical Outlier Detection | Identifies observations that deviate significantly from the statistical characteristics of the data distribution. | a. does not require fraud labels;<br>b. robust methods (e.g., Modified Z-Score) handle skewed distributions and extreme observations.                                      | a. may generate false positives when operational variability is naturally high;<br>b. evaluates statistical deviation rather than behavioural relationships |
| Unsupervised Machine Learning | Learns normal behavioural patterns directly from unlabelled data and identifies multidimensional anomalies.       | a. capable of detecting complex nonlinear relationships;<br>b. adaptive to large-scale data;<br>c. does not require labelled fraud cases.                                  | c. lower interpretability;<br>d. sensitive to parameter selection and feature engineering;<br>e. higher computational complexity.                           |

Building upon these perspectives, the present study adopts an integrated theoretical foundation in which rule-based knowledge, robust statistical analysis, and unsupervised machine learning in anomaly detection. Three methods have been joined to inform the proposed fraud risk assessment framework.

## METHODS

### 1. Research Design

Figure 1 presented overall workflow in this study, which begin with data acquisition to development of the composite fraud risk screening index. The research process consists of four major stages. The first phase involves data collection and preparation to ensure data quality for subsequent analysis, including data cleaning, exploratory, and feature engineering to obtain new variable based on available data. The second phase focuses on anomaly detection by integrating three complementary analytical perspectives: rule based, statistical analysis, and machine learning. These approaches provide complementary evidence for identifying potential fraudulent behaviours. Furthermore, the third phase integrates the previous anomaly indicators into single composite fraud risk screening index by using a weighted aggregation approach. Finally, the results of composite index on each port are classified into five categories to interpret and support risk-based prioritization.

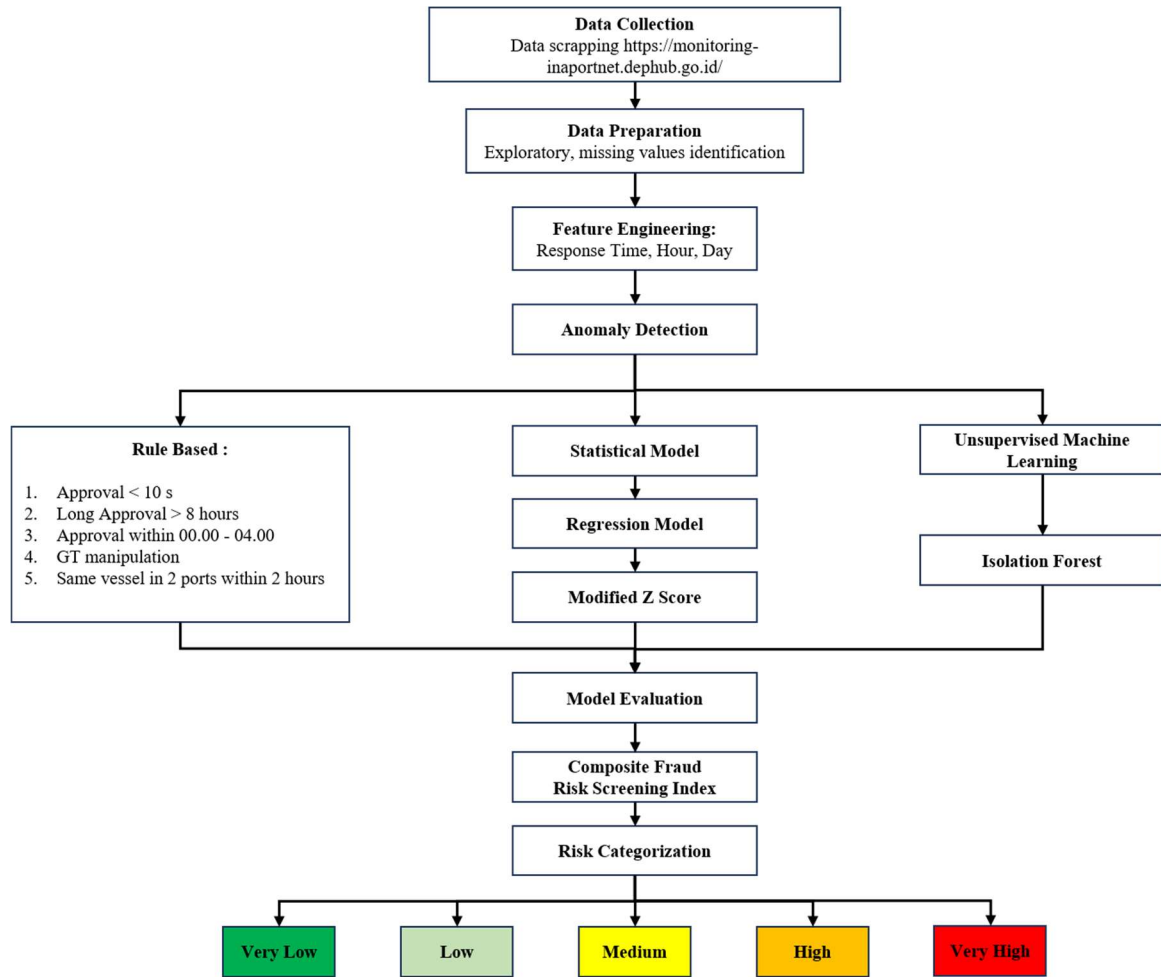


Figure 1 – Research Flowchart Design

## 2. Data source and preparation

The empirical analysis in this study utilizes administrative transaction data of vessel arrival authorization (PKK), extracted from the inaportnet systems. The dataset consists of all PKK transactions processed during 2025 observation period, covering 257 commercial ports across Indonesia and total 746,896 approval transaction.

Prior to subsequent analysis on anomaly detection, preparation procedures were conducted to improve data quality and ensure the reliability. Firstly, exploratory data analysis examines the overall characteristics of dataset, including distribution, temporal pattern, missing values, and potential data quality issues. Secondly, data cleaning assessed missing values, duplicate records, and inconsistent data entries. Following data cleaning, feature engineering was performed to derive additional variables that represents operational characteristics of the approval process such as response time, approval hour and service day. These derived variables complemented the initial dataset providing the analytical foundation for identifying anomalous operational behaviour.

Data preprocessing stage resulted in 12,274 transactions being excluded from the analysis due to scope-related issues. These exclusions consisted of transaction where the submission occurred where the submission occurred prior to 2025 while approval occurred in 2025, and transactions where both submission and approval occurred prior to 2025 but were included in the 2025 extract due to an estimated time of departure (ETD) falling within 2025. Because this study is limited to 2025, datasets concerning PKK submission and approval activities

from other periods are excluded from the analysis. Therefore, a total of 734,622 transactions were retained for further analysis.

Furthermore, it should be noted that during data extraction process from the inaportnet system, a retrieval failure occurred at the following ports as shown in Table 2. As a preliminary assessment of potential missing data bias, comparison between retained and excluded observations on descriptive statistics of approval process was conducted as detailed in Table 3. Relatively, compared dataset resulted small differences range 0% - 1.9%, while minimum and maximum were identical in both groups, the difference of standard deviation showed a relatively small at 1.92%. Therefore, this assessment showed that the retrieval failures at these seven ports did not materially affect the overall distribution characteristics of the dataset, supporting the decision to retain these ports in the subsequent analysis.

Table 2 – data retrieval failures during system extraction

| No | port code | port                  | Month failure | Month Succeed                        | Data Transaction Success |
|----|-----------|-----------------------|---------------|--------------------------------------|--------------------------|
| 1  | IDBDJ     | Banjarmasin           | 12 (domestic) | 12 (international)                   | 1,703                    |
| 2  | IDBOA     | Benoa                 | 12 (domestic) | 12 (international)                   | 87                       |
| 3  | IDBPN     | Balikpapan            | 10 (domestic) | 2 (domestic),<br>12 (international)  | 5,237                    |
| 4  | IDBTN     | Banten                | 1 (domestic)  | 11 (domestic),<br>12 (international) | 12,710                   |
| 5  | IDLBO     | Labuan Bajo           | 8 (domestic)  | 4 (domestic),<br>12 (international)  | 5,271                    |
| 6  | IDSRI     | Samarinda             | 12 (domestic) | 12 (international)                   | 954                      |
| 7  | IDTJB     | Tanjung Balai Karimun | 12 (domestic) | 12 (international)                   | 2,446                    |

Table 3 – Preliminary assessment: retained vs exclude

| Statistics     | Retained data failure | Exclude data failure | Relative differences |
|----------------|-----------------------|----------------------|----------------------|
| Number of Data | 734,622               | 706,214              | 3.87%                |
| Min            | 0.00                  | 0.00                 | 0.00%                |
| Q1             | 2.73                  | 2.72                 | 0.37%                |
| Mean           | 34.57                 | 34.65                | 0.23%                |
| Median         | 7.28                  | 7.25                 | 0.41%                |
| Q3             | 23.02                 | 23.08                | 0.26%                |
| Maximum        | 292,161.55            | 292,161.55           | 0.00%                |
| Std Dev        | 517.82                | 527.76               | 1.92%                |

### 3. Analytical framework

#### a. Rule based

Anomaly detection employed a rule-based approach to identify operational behaviours that explicitly violated predefined business rules on approval process. Based on government regulation issued by Ministry of Transportation (2022) mentioned in PM 8 Tahun 2022, showed that approval process based on document verification is completed within 30 minutes. During that period, officers are given the opportunity to evaluate and verify the completeness and validity of the documents. Therefore, rule-based detection was designed to capture anomalies including as follows:

Table 4 - Justification of rule-based threshold

| No | Rule-base                    | Threshold                                                          | Operational/Legal Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Quick approval               | <10 seconds                                                        | Based on regulation PM 93 Tahun 2013 (Ministry of Transportation, 2013), on the process vessel arrival notification (PKK), the officer should verify several documents including operational, agency, vessel, seaworthiness, cargo manifest, etc. Therefore, quick processes lead to inadequate verification contributing fraud risk (Landini & Spears, 2025)                                                                                                                                                        |
| 2  | Long duration                | >8 hours                                                           | Long approval durations can create opportunities for bribery as payments to accelerate transactions through delays (Warner, 2012). Long duration approval is specifically defined as an approval process taking more than 8 hours, following the legal working hour standards (Indonesian Government, 2020)                                                                                                                                                                                                          |
| 3  | Low Oversight                | 00.00 – 04.00                                                      | Limited oversight of approval processes during outside standard operating hours creates a strategic loophole for fraudster. This strategy is frequently used to divert public and supervisor attention (Cheng, Palmon, Yang, & Yin, 2023)                                                                                                                                                                                                                                                                            |
| 4  | GT manipulation              | Modification of a vessel's Gross Tonnage (GT) data within 6 months | Discrepancies between the gross tonnage (GT) in documents and actual dimensions serve as an indicator of markdown manipulation (Fadly, Krisnafi, & Soeboer, 2020). Administrative process for issuing vessel documents following a change in Gross Tonnage (GT) including submission, registration, document verification, appointment of a surveyor, measurement, validation and document issued (Tedi, Dirhamsyah, & Yuna, 2024). Consequently, updating the vessel's data requires a considerable amount of time. |
| 5  | Same vessel within two ports | < 2 hours                                                          | Based on government regulation PP Nomor 61 Tahun 2009, stated that the optimal distance between ports ranges from 5 to 20 nautical miles. For instance, Ketapang-Gilimanuk port route spans a distance of just 5 nautical miles and standard round trip time of 3 hours or 180 minutes (Tiara, Nugroho, & Setiawan, 2024). Therefore, same vessel across two different ports flags an anomalous operational state, signaling as red flag.                                                                            |

b. Modified Z-score (statistical method)

The second analytical analysis applied statistical anomaly detection based on modified z-score calculated from regression residuals. First, this approach estimated the expected approval process through regression modelling. Approval duration is naturally influenced by port/service volume, complexity, and time factor. Regression model to represent predicted approval duration as follows:

$$T_i = \beta_0 + \beta_1 V + \beta_2 GT + \beta_3 D + \beta_4 H$$

Where, T = predicted approval duration  
V = total service volume in a day in each port  
GT = gross tonnage as complexity variables  
D, H = Day and Hour as time factor

After estimating the regression model, standardized residual anomalies were quantified using the Modified Z-Score based on Median Absolute Deviation (MAD). Therefore, Modified z score:

$$Z_i = 0.6745 \frac{r_i - \bar{r}}{MAD}$$

c. Isolation Forest (Unsupervised machine learning)

This approach was employed to identify multidimensional anomalies that could not be adequately captured through previous analysis (rule based and statistical). The algorithm generated an anomaly score for each transaction, reflecting its relative degree of isolation within the multidimensional feature space. Model was trained on all datasets based on approval duration, vessel capacity in GT, service volume, and time approval. The anomaly outputs generated from these three analytical components were subsequently standardized and integrated into the Composite Fraud Risk Index

The resulting anomaly indicators then were integrated into a composite fraud risk index at port level. Prior to aggregation, three indicators generated from the rule based, modified z-score, and isolation methods were transformed using Min-Max normalization. To ensure comparability among three analytical indicators, the normalized value was ranged from 0.1 to 1.0 in order to avoid assigning a value of absolute zero to any port. This decision reflects that no port should be interpreted as having zero fraud risk exposure and ensure the composite score retains a residual risk floor across all ports.

In addition, an equal-weighting scheme was subsequently applied to the three normalized sub-indices. This approach reflects a conservative assumption that no single detection method is presumed more informative than another prior to case-level validation.

4. Fraud Risk Categorization

According to government regulation on risk management, risk levels are classified into 5 category (Ministry of Transportation, 2023). Therefore, each composite fraud risk screening index port generated in the previous analysis are classified into five ordinal fraud risk level by using a percentile-based approach, consistent with the five-tier risk classification structure Very Low, Low, Medium, High, and Very High. These categories indicate the relative level of operational anomaly observed within each port based on the integrated analytical framework. Alternatively, fixed scale was adopted to classify the categories into five groups as detailed below:

Table 5 – Fix scale composite index category

| Category  | Interval    |
|-----------|-------------|
| Very Low  | 0.10 – 0.28 |
| Low       | 0.28 – 0.46 |
| Medium    | 0.46 – 0.64 |
| High      | 0.64 – 0.82 |
| Very High | 0.82 – 1.00 |

## RESULTS

1. Descriptive Statistics

After data preparation and cleaning, there were 734,622 approval transaction of vessel arrival across all Indonesian ports (257 ports). Table 2 summarizes the empirical distribution of administrative approval durations. The overall mean processing time is 34.57 minutes, while the median is significantly lower at 7.28 minutes. These statistics indicates that most transactions are resolved rapidly. Nevertheless, the presence of extreme outliers reaching a maximum approval duration of 292,161 minutes or more than 200 days indicates that there



were bottlenecks systems on this situation. In contrast, a minimum duration of zero minutes suggests an automated or instant 'rubber-stamping' process. From a service level perspective, approximately 78.82% of PKK approval transactions were completed within the national 30-minute Service Level Agreement. This finding shows that despite the high compliance rate, a significant portion still experienced processing delays more than 30 minutes. In terms on port transaction volume, the service quantity among all ports demonstrates substantial heterogeneity with transaction volumes range from only 2 to 22,240 transactions during 2025. A mean of 2,858.45 and median of 1,229 transactions, indicates a right skewed distribution in which a relatively small number of ports process substantially larger transaction volumes than the majority of ports.

Table 6 – Descriptive statistics of approval duration process in minutes and port volume

| Variable          | Number of Data | Min  | Q1   | Median | Q3    | Max        | Mean    | Std Dev  |
|-------------------|----------------|------|------|--------|-------|------------|---------|----------|
| Approval duration | 736,253        | 0.00 | 2.72 | 7.28   | 23.02 | 292,161.55 | 34.57   | 518.82   |
| Port Volume       | 257            | 2    | 586  | 1,229  | 3,372 | 22,240     | 2,858.5 | 3,922.52 |

## 2. Anomaly Detection

### a. Rule based

The rule-based anomaly detection method identified 56,306 approvals, which represents 7.66% of all 734,622 evaluated PKK approval transaction. These transactions met one or more criteria of established red flag conditions derived from predefined operational rules. Among the identified condition, same-vessel activity across two ports within 2 hours constituted the largest proportion, accounting for 24,367 transactions (3.32%) of the total approval dataset, followed by GT manipulation with 12,926 transactions (1.76%). These findings emphasized that cross port transactional relationships may provide an important dimension for fraud risk screening that cannot be captured by individual transaction records in isolation. Furthermore, approvals occurring within 00:00–04:00 experienced with 11,860 transactions. While unusually long approval duration exhibited around 8,420 approval, short approval duration below 10 seconds reached at 1,078 or 0.15% of total approval. Given these data, this method does not constitute evidence of fraudulent activity, they represent early warning sign for transaction that deviate from expected operational condition and suggests for further examination.

At the port level, Figure 2 illustrates ports with relatively low transaction volumes can exhibit high red flag proportion at 57.01%. This scatter plot identifies the absolute number of flagged transactions and the relative proportion of flagged transactions provide different perspectives on fraud risk. In terms of total red flag transaction, Labuhan port experienced the highest red flag with 4,429 approvals followed by Benete port with 4,349 transactions. As presented in the Table 3, although Lahewa port has a relatively small transaction at 107, this port recorded the highest red flag proportion at 57.01%, followed by Benete at 40.02% and Labuhan as the largest red flag transaction reach at 39.70%. These findings reinforce the importance of distinguishing between frequency, proportion, and type of anomaly in port-level fraud risk assessment for strategic intervention. Ports with a small transaction may produce unstable or disproportionately high percentages when only a limited number of transactions are flagged, while high-volume ports may accumulate a large absolute number of red flags despite having a relatively low proportion.

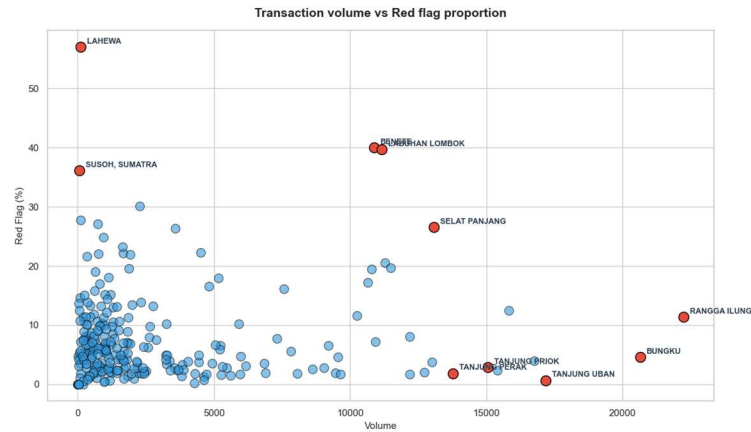


Figure 2 – Red flag percentage on volume transaction across all ports

Table 7 - Top 5 port of red flag percentage

| No | Port    | Volume | Red Flag       |               |                               |                 |                        |       | Percentage of red flags |
|----|---------|--------|----------------|---------------|-------------------------------|-----------------|------------------------|-------|-------------------------|
|    |         |        | Short duration | Long duration | Approval within 00.00 - 04.00 | GT Manipulation | Same vessel in 2 ports | Total |                         |
| 1  | LAHEWA  | 107    | 0              | 9             | 10                            | 33              | 9                      | 61    | 57.01                   |
| 2  | BENETE  | 10867  | 1              | 33            | 1554                          | 531             | 2230                   | 4349  | 40.02                   |
| 3  | LABUHAN | 11156  | 5              | 0             | 1560                          | 638             | 2226                   | 4429  | 39.70                   |
| 4  | SUSOH   | 72     | 0              | 25            | 1                             | 0               | 0                      | 26    | 36.11                   |
| 5  | RAHA    | 2284   | 0              | 61            | 38                            | 1               | 588                    | 688   | 30.12                   |

#### b. Modified Z-score

Statistical analysis on unusual approval durations first estimated the expected approval duration using regression model. The model incorporated daily approval at the port level, gross tonnage (GT) as the size of complexity on each approval, and approval time indicators. Furthermore, categorical transformation was applied into several variables such as GT, day, and approval variable to capture potentially non-linear differences across operational conditions. The resulting model was ( $F$ -statistic = 2114;  $p < 0.001$ ) and achieved an  $R^2$  of 0.036. Although the relatively low  $R^2$  indicates that the model explains only a modest proportion of the variation in approval duration, the purpose of the regression model in this study is not to maximize prediction accuracy or establish causal relationships between operational factors and approval duration. Rather, the model is used to estimate the expected duration under observed operational conditions. Figure detailed regression model evaluation and coefficient of all independent variables.

To assess the robustness of the model specification, an alternative model was estimated using gross tonnage (GT) as a continuous variable instead of a categorical transformation. The categorical specification yielded a higher explanatory power ( $R^2 = 0.036$ ,  $F = 2,114$ ,  $p < 0.001$ ; adj.  $R^2 = 0.036$ ) compared to the continuous specification ( $R^2 = 0.024$ ,  $F = 1,654$ ,  $p < 0.001$ ; adj.  $R^2 = 0.024$ ), as detailed in Figure 3. This improvement suggests that the relationship between GT and approval duration is better captured through categorical thresholds than through a linear functional form, supporting the use of the categorical GT specification in the main model. Furthermore, multicollinearity among the independent variables was assessed using the Variance Inflation Factor (VIF). All

variables in the categorical model returned VIF values below 2 (ranging from 1.03 to 1.73), indicating no evidence of problematic multicollinearity among the predictors.

| OLS Regression Results |                      |                     |             |       |        |        |
|------------------------|----------------------|---------------------|-------------|-------|--------|--------|
| Dep. Variable:         | log_approval_minutes | R-squared:          | 0.836       |       |        |        |
| Model:                 | OLS                  | Adj. R-squared:     | 0.836       |       |        |        |
| Method:                | Least Squares        | F-statistic:        | 2114.       |       |        |        |
| Date:                  | Tue, 01 Sep 2026     | Prob (F-statistic): | 0.00        |       |        |        |
| Time:                  | 13:37:34             | Log-Likelihood:     | -1.2465e+06 |       |        |        |
| No. Observations:      | 734622               | AIC:                | 2.493e+06   |       |        |        |
| DF Residuals:          | 734608               | BIC:                | 2.493e+06   |       |        |        |
| DF Model:              | 13                   |                     |             |       |        |        |
| Covariance Type:       | nonrobust            |                     |             |       |        |        |
|                        | coef                 | std err             | t           | P> t  | [0.025 | 0.975] |
| const                  | 2.1273               | 0.006               | 372.904     | 0.000 | 2.116  | 2.138  |
| daily_approval         | 0.0031               | 7.83e-05            | 39.245      | 0.000 | 0.003  | 0.003  |
| gt                     | -0.0517              | 0.006               | -9.094      | 0.000 | -0.063 | -0.041 |
| day_Monday             | -0.0055              | 0.006               | -0.946      | 0.344 | -0.017 | 0.006  |
| day_Saturday           | 0.0042               | 0.006               | 0.707       | 0.479 | -0.007 | 0.016  |
| day_Sunday             | -0.0221              | 0.006               | -3.874      | 0.000 | -0.033 | -0.011 |
| day_Thursday           | -0.0353              | 0.006               | -6.194      | 0.000 | -0.046 | -0.024 |
| day_Tuesday            | -0.0174              | 0.006               | -3.044      | 0.002 | -0.029 | -0.006 |
| day_Wednesday          | 0.2564               | 0.004               | 58.617      | 0.000 | 0.248  | 0.265  |
| gt_category_Q2         | -0.0959              | 0.004               | -21.920     | 0.000 | -0.105 | -0.087 |
| gt_category_Q3         | 0.2312               | 0.004               | 52.602      | 0.000 | 0.223  | 0.240  |

(a)

| OLS Regression Results |                      |                     |             |       |          |          |
|------------------------|----------------------|---------------------|-------------|-------|----------|----------|
| Dep. Variable:         | log_approval_minutes | R-squared:          | 0.824       |       |          |          |
| Model:                 | OLS                  | Adj. R-squared:     | 0.824       |       |          |          |
| Method:                | Least Squares        | F-statistic:        | 1654.       |       |          |          |
| Date:                  | Tue, 01 Sep 2026     | Prob (F-statistic): | 0.00        |       |          |          |
| Time:                  | 13:57:25             | Log-Likelihood:     | -1.2510e+06 |       |          |          |
| No. Observations:      | 734622               | AIC:                | 2.502e+06   |       |          |          |
| DF Residuals:          | 734618               | BIC:                | 2.502e+06   |       |          |          |
| DF Model:              | 11                   |                     |             |       |          |          |
| Covariance Type:       | nonrobust            |                     |             |       |          |          |
|                        | coef                 | std err             | t           | P> t  | [0.025   | 0.975]   |
| const                  | 2.1917               | 0.005               | 421.328     | 0.000 | 2.182    | 2.202    |
| daily_approval         | 0.0042               | 7.78e-05            | 53.738      | 0.000 | 0.004    | 0.004    |
| gt                     | 2.722e-06            | 1.96e-07            | 13.904      | 0.000 | 2.34e-06 | 3.11e-06 |
| day_Monday             | -0.0513              | 0.006               | -9.969      | 0.000 | -0.063   | -0.040   |
| day_Saturday           | -0.0061              | 0.006               | -1.048      | 0.295 | -0.018   | 0.005    |
| day_Sunday             | 0.0055               | 0.006               | 0.927       | 0.354 | -0.006   | 0.017    |
| day_Thursday           | -0.0231              | 0.006               | -4.022      | 0.000 | -0.034   | -0.012   |
| day_Tuesday            | -0.0353              | 0.006               | -6.168      | 0.000 | -0.047   | -0.024   |
| day_Wednesday          | -0.0150              | 0.006               | -2.621      | 0.009 | -0.026   | -0.004   |
| hour_cat_Evening       | 0.1202               | 0.004               | 29.687      | 0.000 | 0.112    | 0.128    |
| hour_cat_Morning       | -0.0592              | 0.004               | -15.014     | 0.000 | -0.067   | -0.051   |

(b)

(a)

(b)

Figure 3 - regression model evaluation: (a) GT as categorical, (b) GT as numerical

By using a lower tail threshold of Z scores = -2.5, this analysis has succeeded to identify 465 transactions exhibiting Modified Z-Scores below the predefined threshold. these observations represent approximately 0.06% of relatively to 734,622 transactions included in the regression analysis, detailed of port with total number of anomaly transaction as shown in Table 4. In terms of port level, the lowest mean modified z-score was approximately -0.6, which remain above the threshold. Therefore, this finding highlight that although there were individual transactions exhibited statistically extreme deviations, these observations were not sufficiently concentrated within any single port to shift its overall mean anomaly score beyond the predefined threshold.

Table 8 – List of port with anomaly transaction based on modified z-score

| No | Port          | Number of anomaly transaction |
|----|---------------|-------------------------------|
| 1  | RANGGA ILUNG  | 274                           |
| 2  | BAKAUHENI     | 135                           |
| 3  | TANJUNG PERAK | 15                            |
| 4  | BUNGKU        | 14                            |
| 5  | WEDA          | 14                            |
| 6  | TANJUNG PRIOK | 5                             |
| 7  | BAU-BAU       | 4                             |
| 8  | PALEMBANG     | 3                             |
| 9  | BALIKPAPAN    | 1                             |
|    | Total         | 465                           |

#### c. Isolation forest

The last analytical method to identify anomaly transaction utilised isolation forest algorithm. This analysis incorporated 3 main features, including approval duration in minutes, gross tonnage, and approval time (day, hour). Because GT and approval duration differ substantially in scale from the hour and day indicators, all numerical features were standardized using z-score normalization prior to model fitting, to prevent features with larger numerical ranges from disproportionately influencing the isolation process.

The model was configured with 100 estimators, a maximum sample size of 256 observations per tree ('auto', the default setting), and full feature usage per split (max\_features = 1.0). The contamination parameter, which specifies the expected proportion of anomalies in the dataset, was set to 0.07. It is consistent with the proportion of transactions flagged under the rule-based detection criteria. A fixed random seed (random\_state = 42) was applied to ensure reproducibility of results across runs.

At the transaction level, the resulting isolation forest anomaly scores ranged approximately from -0.36 to -0.75, transaction with scores closer to zero indicates that approval condition remains more consistent with the prevailing population, while increasing negative scores represent unusual approval behaviour. Meantime, at the port level, aggregated transaction level calculates the mean anomaly score, which ranged around -0.41 to -0.60. Samarinda port experiences the lowest score of isolation forest, which means that several transactions have potentially identified as anomaly approval process.

### 3. Composite Fraud Risk Screening Index (CFRSI) and Classification

The resulted fraud risk index demonstrates substantial variation across 257 Indonesian ports. The top 5 ports of composites scores as shown in the Table 5. Lahewa port recorded the highest composite index at approximately 0.77, followed by Samarinda with 0.68 and Pomako at 0.66. Among the top five ports, Benete also recorded a comparatively large transaction volume with 10,867 approvals. Given this dataset, a high composite index may not correlate with high every individual anomaly indicator. However, this is because anomalous signals from one of analytical approaches accumulate to produce a relatively elevated overall risk score. To facilitate risk-based prioritization, the continuous CFRSI were classified into five ordinal risk categories with a percentile-based approach utilized to distribute into five relative risk groups.

Furthermore, Table 6 presents the distribution of ports across the five risk categories under both percentile and fixed approaches. In percentile approach, ports are distributed in equal proportions across categories (51-52 ports within each category). However, under the fixed-scale classification which applies equal interval range of the composite score, the distribution differs substantially. The majority of ports fall within Low category with 185 ports, followed by medium category at 48, and very low at 21 ports. Only 3 ports fall within the High category, and no ports meet the threshold for the Very High category. Given this dataset, It confirms that the Very High classification under the percentile-based approach identifies ports with the highest composite scores relative to other Indonesian ports in this dataset, rather than ports exhibiting an absolute or extreme level of fraud risk.

It is important to emphasize that the resulting risk classification represent indicators of anomalous operational patterns within the approval process, rather than confirmed of fraud. A high composite score reflects should be interpreted as a red flag requiring further examination, not as evidence of fraudulent activity. Empirical validation of the value of CFRSI against fraud cases, audit findings, or expert reviews was beyond the current study. Therefore, this highlight is an essential direction for future research.

Table 9 - Top 5 composite fraud risk scoring index

| No | Port      | Volume | Anomaly Detection Approach |                   |                        | Composite Index |
|----|-----------|--------|----------------------------|-------------------|------------------------|-----------------|
|    |           |        | Rule Based Index           | Statistical Index | Machine Learning Index |                 |
| 1  | Lahewa    | 107    | 1.00                       | 0.86              | 0.45                   | 0.77            |
| 2  | Samarinda | 954    | 0.28                       | 0.76              | 1.00                   | 0.68            |
| 3  | Pomako    | 756    | 0.45                       | 0.83              | 0.69                   | 0.66            |
| 4  | Sinabang  | 351    | 0.44                       | 0.95              | 0.40                   | 0.60            |
| 5  | Benete    | 10867  | 0.73                       | 0.74              | 0.29                   | 0.59            |

Table 10 - Classification of composite index based on percentile approach and fixed scale

| No | Risk Category | Percentile approach |                | Fixed-scale |                |
|----|---------------|---------------------|----------------|-------------|----------------|
|    |               | Range               | Number of Port | Range       | Number of Port |
| 1  | Very High     | 0.458 - 0.771       | 52             | 0.82 – 1.00 | 0              |
| 2  | High          | 0.413 - 0.457       | 51             | 0.64 – 0.82 | 3              |
| 3  | Medium        | 0.381 - 0.412       | 51             | 0.46 – 0.64 | 48             |
| 4  | Low           | 0.352 - 0.380       | 51             | 0.28 – 0.46 | 185            |
| 5  | Very Low      | 0.125 - 0.351       | 52             | 0.10 – 0.28 | 21             |

## DISCUSSION

This study aimed to develop a data-driven framework for detecting potentially anomalous transaction within the approval process of vessel arrival in inaportnet application. Through composite fraud risk screening index, this study enables to compare and classify ports into five relative fraud risk levels. Beyond its analytical function, the practical value of the CFRSI depends on its integration into a formal fraud risk management (FRM) process with clear governance arrangements. Adopting financial institution regulation (OJK, 2024), the implementation of anti-fraud strategy consists of 4 pillars including prevention, detection, investigation, and monitoring evaluation. The proposed framework in this study aims as an early warning detection mechanism, generating relative risk alerts based on aggregated anomaly indicators rather than confirmed finding of fraudulent activity. Then, internal audit or anti-fraud unit acts to receive and follow up the alerts. An escalation protocol would determine how alerts of differing severity are routed, such as port-level review for moderate alerts and centralized escalation for the highest-priority cases. Flagged ports would then proceed to the investigation stage, involving evidence gathering and case-level review of the underlying transactions within formal audit or law enforcement processes. When investigation does not account for fraudulent activity, a structured false-positive handling mechanism should document the findings and fed back into the model to inform recalibration of thresholds and assumptions.

Sustaining the reliability of the CFRSI over time also requires formal model governance, including defined authority over parameter changes (e.g., rule-based thresholds, contamination assumptions, weighting schemes) and periodic model review as operational patterns evolve. To ensure accountability and traceability of the FRM process, documentation of audit trail is essential including red-flag generation, escalation, investigation outcome, and case closure. Given that the underlying data involves the operational records of vessels, agents, and port operators who may not ultimately be implicated in wrongdoing, appropriate access controls and data privacy safeguards should restrict visibility of port-level or case-level risk scores to authorized personnel involved in the FRM process. Finally, management-override controls should allow for documented exceptions. For instance, where operational context justifies deviation from an automated alert, while ensuring that such overrides are themselves logged and subject to audit, to prevent the override mechanism from becoming a vector for circumventing the detection system.

Apart from benefits derived from this research, several limitations should be acknowledged. Firstly, inaportnet supports multiple types of administrative processes within maritime sector, the scope of the analysis is limited to PKK approval transactions. While other services such as

sailing approval loading and unloading activity planning, or other digital workflows, were not included in the present analysis. Consequently, the resulting risk profile should be interpreted specifically within the PKK approval process. Secondly, the study is conducted based on transaction data during 2025. A single-year observation period may not fully capture temporal variations in operational behaviour. Thirdly, the completeness of the empirical dataset was influenced by technical limitations in the data collection process. Some transactions could not be retrieved because of server-side errors.

Furthermore, to advance the data-driven of anti-fraud governance, future research should explore on the three primary dimensions:

1. Field validation, the composite index derived as an early warning indicator. High risk scores demonstrate some significant operational anomalies. Future studies should validate and conduct targeted field audits.
2. Strategic action through quadrant analysis incorporating service volume to translate risk scores into intervention strategies
3. Methodological scalability across other inaportnet modules. The analytical architecture design in this study, combine rule engines, statistics, and unsupervised machine learning. This approach potentially applicable to other administrative processes characterized by high transaction volumes, heterogeneous operational units, limited fraud labels, and digitally recorded process histories.

## CONCLUSION

This study established a data-driven fraud risk screening framework and constructed a Composite Fraud Risk Screening Index (CFRSI) to support early stage of risk identification in the approval process of vessel arrival (PKK) within inaportnet system. By conceptualizing fraud risk screening as early warning mechanism, this research addresses the need for scalable, automated risk screening within high volume of digital government platforms. Three complementary analytical approaches were integrated to construct the index including rule-based, statistical, and unsupervised machine learning. The driven rule engine has identified 56,306 transactions (7.98% of total dataset), while the regression-based Modified Z-Score isolated 465 severe processing duration outliers, and the Isolation Forest algorithm uncovered non-linear, multivariate deviations. These three indicators were aggregated into port level CFRSI, which was used to categorize ports across five distinct risk tiers (Very Low to Very High). Furthermore, three indicators represent operational red flags indicative of patterns warranting further scrutiny, not confirmed instances of fraud. Accordingly, the CFRSI's contribution should be understood strictly within the scope of risk-based prioritization: it offers a structured, scalable mechanism to help supervisory bodies allocate limited audit and investigative resources. Future research should focus on validating identified red flags through verified audit field records, assessing alternative index parameter, and expanding the model to other digital maritime workflows for strategic anti-fraud governance.

## REFERENCES

- ACFE. (2026, 06/08/2026). Fraud 101: What Is Fraud? Retrieved from <https://www.acfe.com/fraud-resources/fraud-101-what-is-fraud>
- Admin, P. (2018, 2018/06/29). Kemenhub Luncurkan Aplikasi Inaportnet 2.0 dan DO Online di Lima Pelabuhan. Retrieved from <https://hubla.dephub.go.id/home/post/read/5603/kemenhub-luncurkan-aplikasi-inaportnet-2-0-dan-do-online-di-lima-pelabuhan>

- Alfian, A., Ritchi, H., & Adrianto, Z. (2023). Fraud analytics practices in public-sector transactions: a systematic review. *Journal of Public Budgeting, Accounting and Financial Management*, 35(5), 685-710. doi:10.1108/JPBAFM-11-2022-0175
- Alghushairy, O., Alsini, R., Soule, T., & Ma, X. (2021). A review of local outlier factor algorithms for outlier detection in big data streams. *Big Data and Cognitive Computing*, 5(1), 1-24. doi:10.3390/bdcc5010001
- Catelani, M., Ciani, L., Alfonso, C. G., Grasso, F., Paolucci, L., & Patrizi, G. (2024). *Introduction to clustering unsupervised machine learning algorithms applied to power quality disturbances*. Paper presented at the 2024 IEEE International Workshop on Metrology for Industry 4.0 and IoT, MetroInd4.0 and IoT 2024 - Proceedings.
- Cheng, X., Palmon, D., Yang, Y., & Yin, C. (2023). Strategic Earnings Announcement Timing and Fraud Detection. *Journal of Business Ethics*, 182(3), 851-874. doi:10.1007/s10551-021-05029-2
- Darban, Z. Z., Webb, G. I., Pan, S., Aggarwal, C. C., & Salehi, M. (2025). CARLA: Self-supervised contrastive representation learning for time series anomaly detection. *Pattern Recognition*, 157. doi:10.1016/j.patcog.2024.110874
- Fadly, Z. R., Krisnafi, Y., & Soeboer, D. A. (2020). Measurement of dimensions and calculation of danish seine fishing vessel volume. *AACL Bioflux*, 13(2), 1003-1015. Retrieved from <https://www.scopus.com/pages/publications/85085307182?origin=resultslist>
- Ghorbani, A. A., Lu, W., & Tavallaee, M. (2010). Theoretical foundation of detection. *Advances in Information Security*, 47, 73-114. doi:10.1007/978-0-387-88771-5\_4
- Indonesian Government. (2020). *Undang-Undang Republik Indonesia Nomor 11 Tahun 2020 tentang Cipta Kerja*. Retrieved from Jakarta: <https://jdih.kemenkoinfra.go.id/id/undang-undang-republik-indonesia-no-11-tahun-2020>
- Kraiem, I. B., Ghazzi, F., Peninou, A., Roman-Jimenez, G., & Teste, O. (2021). *Human-interpretable rules for anomaly detection in time-series*. Paper presented at the Advances in Database Technology - EDBT.
- Kulmie, D. A. (2026). Applying the fraud diamond theory to public financial crime: An empirical assessment of causal factors. *Asian Development Policy Review*, 14(2), 68-81. doi:10.55493/5008.v14i2.6027
- Landini, A., & Spears, R. (2025). Banks approved fraudulent loans to capture origination fees: evidence from the Paycheck Protection Program. *Journal of Financial Crime*, 32(4), 763-775. doi:10.1108/JFC-07-2024-0201
- Lu, C., Zhao, Y., Xing, K., & Liu, J. (2025). Digital transformation and corporate fraud: Evidence from China. *International Review of Financial Analysis*, 106. doi:10.1016/j.irfa.2025.104476
- Ministry of Transportation. (2013). *Peraturan Menteri Perhubungan Nomor PM 93 Tahun 2013 tentang Penyelenggaraan dan Pengusahaan Angkutan Laut*. Retrieved from Jakarta: <https://jdih.dephub.go.id/peraturan/detail?data=55VcaGas8lQJ1YPoKJ4dOi8hdnzWYt1fV4qDMZuUiKZd8m4NN5l83Bo4TnRGw5vpNG4JAtpvyrnwl8W2SD2Gmuv24aChn2tqCFH48eKUgb6PIH9eCRrCrvqqf6fGCSSBlt7aCaFSxO1L0rX8VI1DBbn4KT>
- Ministry of Transportation. (2022). PM 8 Tahun 2022. In. Jakarta: Kementerian Perhubungan Republik Indonesia.
- Ministry of Transportation. (2023). Keputusan Menteri Perhubungan Republik Indonesia Nomor KM 69 Tahun 2023 tentang Manajemen Risiko di Lingkungan Kementerian Perhubungan. In. Jakarta, Indonesia: Kementerian Perhubungan Republik Indonesia.
- OJK. (2024). *Peraturan Otoritas Jasa Keuangan Republik Indonesia Nomor 12 Tahun 2024 tentang Penerapan Strategi Anti Fraud bagi Lembaga Jasa Keuangan*. Jakarta: Lembaran Negara Republik Indonesia / Peraturan OJK
- Portela, F. G., Almenares Mendoza, F., & Benavides, L. C. (2019). *Evaluation of the performance of supervised and unsupervised Machine learning techniques for intrusion detection*. Paper

- presented at the 2019 IEEE International Conference on Applied Science and Advanced Technology, iCASAT 2019.
- Prusti, D., Kumar, A., Purusottam, I. S., & Rath, S. K. (2021). *A design methodology for web-based services to detect fraudulent transactions in credit card*. Paper presented at the ACM International Conference Proceeding Series.
- Supriadi, T., Tjakrawala, K., Suryadnyana, N. A., Sjam, J. M. E., & Marota, R. (2025). FRAUD PREVENTION IN THE PUBLIC SECTOR: THE ROLE OF INTERNAL AUDIT. *Ikonomicheski Izsledvania*, 34(3), 170-183. Retrieved from <https://www.scopus.com/pages/publications/105002072965?origin=resultslist>
- Tedi, K., Dirhamsyah, D., & Yuna, S. (2024). Prosedur Penerbitan Surat Ukur Kapal Baru Luncur Barge Patria 3222 pada KSOP oleh PT Menara Shipping Agency Batam. *MASMAN Master Manajemen*, 2(4), 126-135. doi:10.59603/masman.v2i4.584
- Tiara, A., Nugroho, B. K., & Setiawan, B. (2024). Evaluasi Kebutuhan Kapal Pada Lintasan Ketapang-Gilimanuk Selama Masa Natal dan Tahun Baru 2024/2025. *IWJ: Inland Waterways Journal*, 6(2). doi:10.54249/iwj.v6i2.225
- Warner, C. M. (2012). Bribery. In *The Wiley-Blackwell Encyclopedia of Globalization* (pp. 1-1).
- Yaro, A. S., Maly, F., Prazak, P., & Maly, K. (2024). Outlier Detection Performance of a Modified Z-Score Method in Time-Series RSS Observation With Hybrid Scale Estimators. *IEEE Access*, 12, 12785-12796. doi:10.1109/ACCESS.2024.3356731